VARDHAN YERRAMSETTI

Python Full Stack Developer • AWS Certified Cloud Practitioner • Data Science

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PROFESSIONAL SUMMARY

Results-driven **Python Full Stack Developer** and **AWS Certified Cloud Practitioner** with specialized expertise in **Data Science, Machine Learning**, and cloud microservices. Proficient in architecting high-throughput backend APIs using Flask and FastAPI, implementing mathematical and machine learning models (BG/NBD, Gamma-Gamma, OpenCV), engineering normalized relational database schemas with **-25% query latency optimizations**, and crafting reactive frontend applications in React.js and Next.js. Backed by 8 production deployments across AWS and Vercel.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), SQL, C++, HTML5, CSS3

Backend & APIs: Flask, FastAPI, Django, RESTful APIs, Serverless, JWT Auth

Data Science & ML: OpenCV, Scikit-learn, BG/NBD, Gamma-Gamma, Plotly 3D WebGL

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, IAM), Vercel, Docker, Git/GitHub

Databases: MySQL, PyMySQL, Relational 3NF, Compound B-Tree Indexing (-25%)

Frontend: React.js 19, Next.js, Tailwind CSS, DOM Manipulation, SPAs

KEY TECHNICAL PROJECTS

Verified Live Deployments & Source Code

Scalable Task & Data Management Cloud Portal

`Python, Flask REST API, React.js, MySQL, AWS EC2/S3, Docker`

Live App: task-cloud-portal.vercel.app • Code: github.com/vardhanyerramsetti71beep/task-cloud-portal

- Architected a responsive multi-tenant cloud portal with strict organization data isolation across tasks, assets, and audit logs.
- Engineered compound B-Tree indexes on `(tenant_id, status)` in MySQL, reducing query execution latency by **25%** (18.4ms to 13.8ms).
- Decoupled binary storage into **Amazon S3** with TLS 1.2+ security policies and containerized execution on **AWS EC2** via Docker.

OmniPulse Enterprise CLV & Customer Retention Prediction Engine

`FastAPI, Scikit-learn, BG/NBD, Gamma-Gamma, Python`

Live App: omni-pulse-enterprise-clv-churn-int-teal.vercel.app • Code: GitHub Repo

- Implemented **Beta-Geometric/Negative Binomial (BG/NBD)** probabilistic models to compute real-time customer vitality probability `P(Alive)`.
- Integrated **Gamma-Gamma** monetary sub-models to forecast 12-month expected customer lifetime value (CLV) with sub-15ms prediction latency.
- Designed high-throughput in-memory REST inference endpoints with Pydantic contract validation for churn risk triage.

Music Recommendation System Based on Facial Expression Recognition

`Python, Flask REST API, OpenCV, Scikit-learn, WebRTC`

Live App: music-recommending.vercel.app • Code: GitHub Repo

- Built an end-to-end computer vision pipeline analyzing facial expressions via live webcam video stream with **>90% classification accuracy**.
- Engineered high-throughput Flask RESTful APIs managing asynchronous model inference and targeted audio playlist dispatch.
- Optimized frame processing pipelines using OpenCV for low-latency facial landmark extraction in production environments.

90-Day Body Transformation Biometric 3D Manifolds Dashboard

`Python, Plotly 3D WebGL, Biometrics, HTML5/CSS3, JavaScript`

Live App: 90-days-body-transformation.vercel.app • Code: GitHub Repo

- Developed a biometric analytics platform converting time-series physical measurements into interactive **Plotly 3D WebGL surface meshes**.
- Implemented mathematical data smoothing algorithms to track morphological variations across circumference, body fat %, and caloric intake timelines.
- Engineered interactive 3D spatial rotation controls and temporal telemetry sliders for exploratory health metrics analysis.

S.V.T. Tailors Bespoke Atelier & 5-Stage Workshop Tracker

`Python Serverless, MySQL, PyMySQL, React.js, Tailwind CSS`

Live App: svt-tailors.vercel.app • Code: GitHub Repo

- Built an end-to-end digital atelier configurator with dynamic parametric garment customizer (fabrics, collar, cuffs, measurements).
- Architected a 5-stage workshop state machine (*Order Logged → Fabric Cut → Master Stitch → Steam Press → Ready*) in MySQL with **<35ms latency**.
- Engineered public tracking portal providing transparent visibility into handcrafted garment fabrication milestones.

MS Pickles Artisanal Andhra E-Commerce Platform `JavaScript (ES6+), HTML5/CSS3, Tailwind CSS, LocalStorage, Vercel`

Live App: **ms-pickles.vercel.app** • Code: **GitHub Repo**

- Engineered a responsive, mobile-first artisanal food e-commerce platform with dynamic flavor filtering and localized regional UX.
- Implemented persistent client-side cart state management supporting real-time price calculations, quantity updates, and local storage hydration.
- Optimized critical rendering paths and asset delivery, achieving sub-second first contentful paint (FCP).

WORK EXPERIENCE

Operations & Logistics Associate — Instakart Services Pvt. Ltd. (Flipkart Logistics) `Jan 2026 – Apr 2026`

- Supervised real-time warehouse operations, order dispatch processing, and inventory databases, maintaining **99% data integrity**.
- Coordinated cross-functional workflows between ground teams and dispatch systems, eliminating bottlenecks and driving a **15% increase in throughput**.
- Conducted root-cause telemetry audits across fulfillment logistics pipelines to minimize order variance and transit latency.

Teaching Assistant (Computer Science) — Sri Srinivasa Gravity Junior College `Jun 2022 – Aug 2022`

- Mentored 40+ engineering aspirants in fundamental computer science principles, Object-Oriented Programming (OOP), and algorithmic problem-solving.
- Conducted weekly practical debugging labs in Python and C++ to build foundational algorithmic fluency.

Direct Marketing Coordinator — Future Generali Total Health Insurance `Aug 2022 – Jan 2023`

- Executed targeted digital campaigns across multi-touch customer channels, engaging 5,000+ prospective leads.
- Analyzed conversion telemetry and customer demographic funnels, driving a 10% uplift in qualified lead acquisition.

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science and Engineering

Jawaharlal Nehru Technological University (JNTU) Kakinada • 2018 – 2026

CGPA: 6.5 / 10.0

Core: Data Structures, Database Management, Operating Systems, Computer Networks

CERTIFICATIONS

- ✓ **AWS Certified Cloud Practitioner** — Amazon Web Services (2023)
- ✓ **AWS Academy Graduate: Cloud Foundations** — AWS (2023)
- ✓ **Android Enterprise Associate** — Google (2022)
- ✓ **Cisco Networking Essentials** — Cisco Networking Academy (2022)